

HOMEWORK 2 FOR MATH 2860 (ELEMENTARY DIFFERENTIAL EQUATIONS)

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Question Q1. Solve the following separable

a) $y' = xe^{x^2}$.

Solution

Separate / integrate:

$$\frac{dy}{dx} = xe^{x^2} \implies y = \int xe^{x^2} dx + C.$$

Use the substitution $u = x^2$, $du = 2x dx$:

$$\int xe^{x^2} dx = \frac{1}{2} \int e^u du = \frac{1}{2} e^u + C = \frac{1}{2} e^{x^2} + C.$$

Thus

$$y(x) = \frac{1}{2} e^{x^2} + C.$$

b) $y' = \frac{\ln(x)}{x^2}$.

Solution

Integrate (note: domain $x > 0$ because of $\ln x$):

$$y = \int \frac{\ln x}{x^2} dx + C.$$

Use integration by parts: let $u = \ln x$, $dv = x^{-2} dx$. Then $du = \frac{1}{x} dx$, $v = -\frac{1}{x}$. Hence

$$\begin{aligned} \int \frac{\ln x}{x^2} dx &= uv - \int v du \\ &= -\frac{\ln x}{x} - \int \left(-\frac{1}{x} \cdot \frac{1}{x}\right) dx \\ &= -\frac{\ln x}{x} + \int \frac{1}{x^2} dx. \end{aligned}$$

Now compute the remaining integral:

$$\int \frac{1}{x^2} dx = -\frac{1}{x},$$

so

$$\int \frac{\ln x}{x^2} dx = -\frac{\ln x}{x} - \frac{1}{x} + C = -\frac{\ln x + 1}{x} + C.$$

Thus

$$y(x) = -\frac{\ln x + 1}{x} + C, \quad x > 0.$$

c) $y' = \sqrt{4 - y^2}$.

Solution

Separate:

$$\frac{dy}{dx} = \sqrt{4 - y^2} \quad \Rightarrow \quad \frac{dy}{\sqrt{4 - y^2}} = dx.$$

Integrate both sides. Use $a = 2$ formula $\int \frac{dy}{\sqrt{a^2 - y^2}} = \arcsin\left(\frac{y}{a}\right) + C$:

$$\arcsin\left(\frac{y}{2}\right) = x + C.$$

Solve for y :

$$\frac{y}{2} = \sin(x + C) \quad \Rightarrow \quad y(x) = 2 \sin(x + C).$$

(Equivalently one may write $y(x) = 2 \sin(x + C)$ with the understanding that the original ODE requires the right-hand side nonnegative, so the solution is valid on intervals where $\cos(x + C) \geq 0$ if one enforces $y' = \sqrt{4 - y^2}$ as the positive square root. The implicit form $\arcsin(y/2) = x + C$ is the safest global statement.)

Thus

$$\arcsin\left(\frac{y}{2}\right) = x + C \quad \text{or} \quad y(x) = 2 \sin(x + C).$$

d) $y' = \frac{\arctan(x)}{1 + x^2}$.

Solution

Separate / integrate:

$$y = \int \frac{\arctan x}{1 + x^2} dx + C.$$

Use substitution $u = \arctan x$. Then $du = \frac{1}{1 + x^2} dx$, so the integral becomes

$$\int u du = \frac{1}{2}u^2 + C = \frac{1}{2}(\arctan x)^2 + C.$$

Therefore

$$y(x) = \frac{1}{2}(\arctan x)^2 + C.$$